

equipment identification (IMEI) number. This provides for a moderate amount of security. In later versions of GSM, the signal from the phone to the base unit can be encrypted, after the phone is authenticated (recognised as genuine) by the base unit.

4.3 CDMA

The other major standard for mobile telephony in use in India, CDMA (Code Division Multiple Access), is the standard used by operators such as Reliance Communications and Tata Teleservices. While not allowing easy interoperability between networks or interchange ability with handsets, some argue that the CDMA standard is still better for India as it allows for much lower power requirements at the base station. The CDMA standard also witnesses very few low coverage problems due to the fact that the range of each cell is very high.

The core concept behind CDMA phones (the standard is more technically known as the IS 95 standard, and is patented by Qualcomm, a US based company), is code division multiplexing. While GSM chooses to divide the transmissions in time from multiple phones in a cell, CDMA transmits all of them at once, encrypts them, breaks them down into coded packets which are transmitted over a wide spectrum of frequencies and reassembled at the base station. The advantages are obvious, whereas GSM cells have a maximum number of phones that can make simultaneous calls (you will presumably often have come across the “network busy” error message on your GSM phone), CDMA phones have no theoretical limits on the number of phones within a cell. Hence, unlike GSM phones, CDMA phones do not have a specific



The SIM Card

An interesting story behind CDMA is the battle between Qualcomm and Ericsson over the technology. Ericsson backed out giving way to Qualcomm ownership over CDMA